5. Audio coding standards

based on Spanias, Painter & Atti, Chapter 10

Key question: how do standard codecs combine transforms, psychoacoustics, quantization and bitstream syntax?

Roadmap

From efficient waveform compression to adaptive, immersive audio

01 · Representations

MIDI · mono/stereo/surround · channel-based audio

02 · Perceptual coding

MPEG standards · MP3 · AAC · MPEG-4 Audio

03 · Modern coding

Opus · xHE-AAC · speech/music · low latency

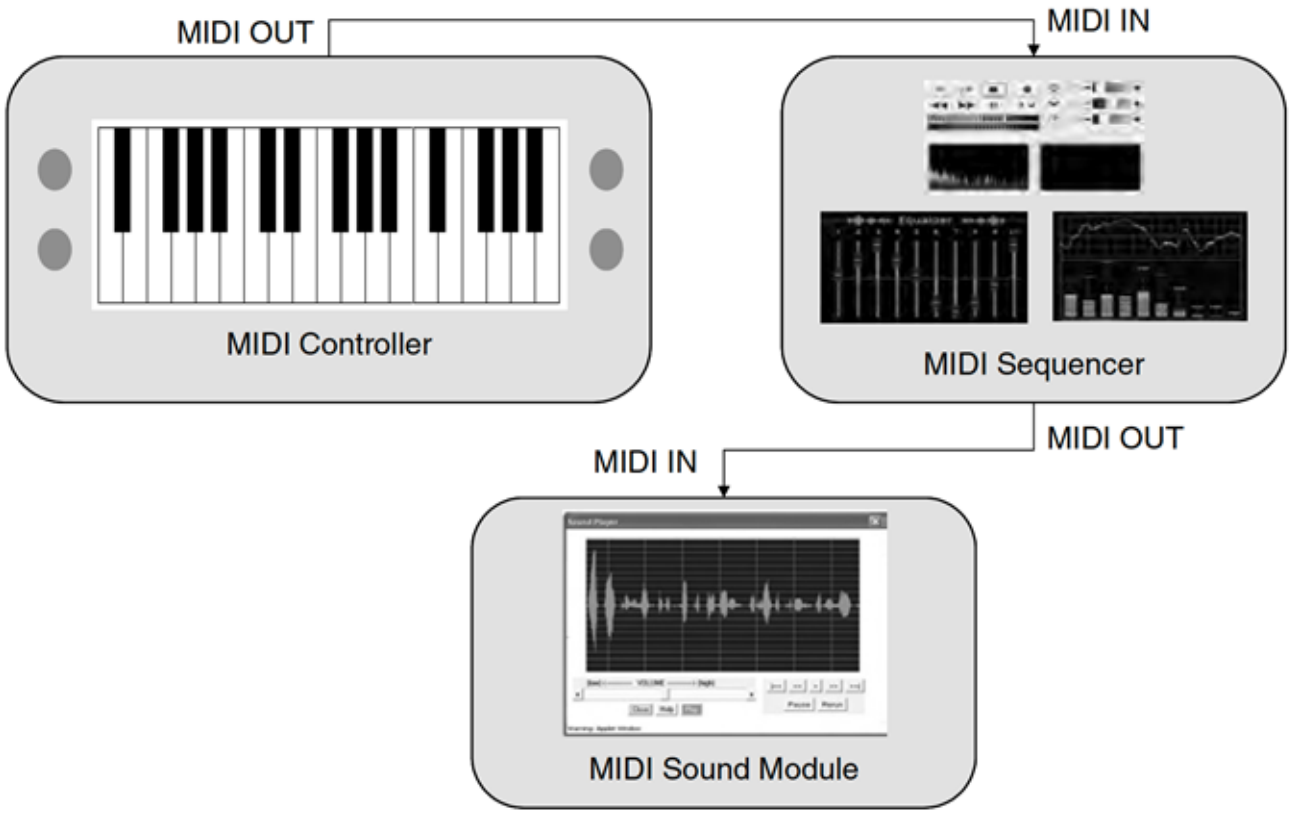
04 · Immersive + lossless

Dolby Atmos · JOC · MPEG-H · FLAC · FFmpeg

Key distinction Compressing waveforms ≠ representing and rendering an audio scene

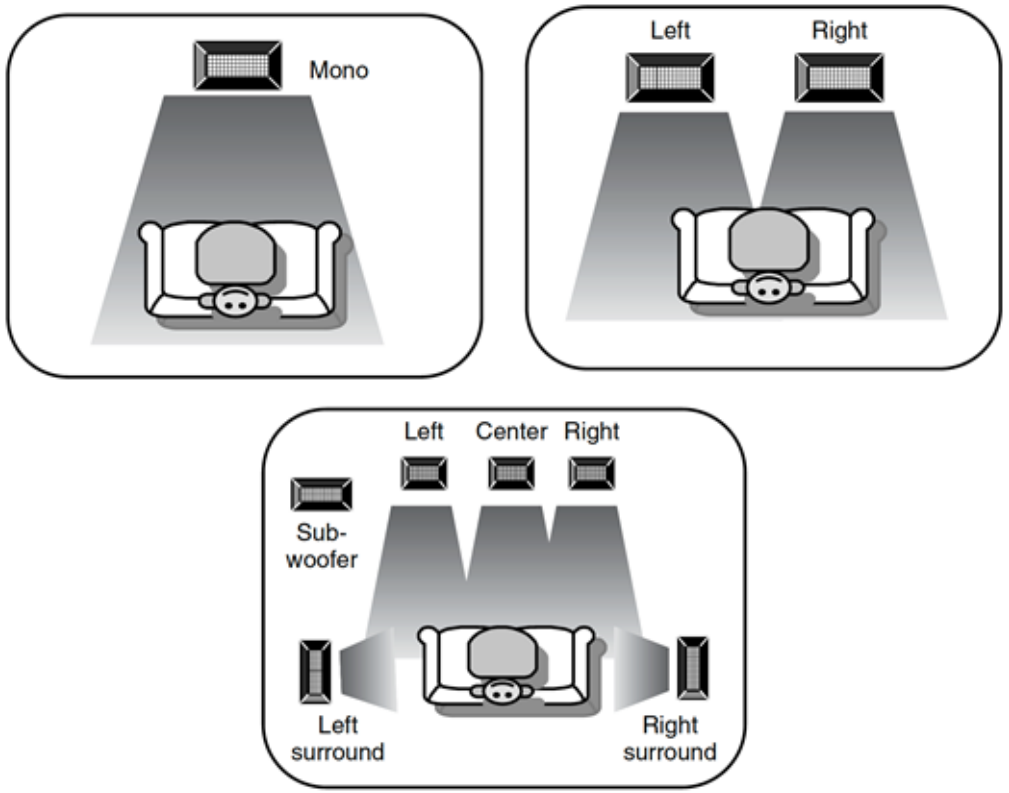
MIDI: symbolic control, not audio samples

MIDI transmits performance events and control messages between devices.



Mono, stereo and surround formats

Channel layout determines how many audio streams and spatial cues must be preserved.



MPEG audio standards: the first generations

From stereo perceptual coding to efficient multichannel audio.

Standard	Main innovation	Why it matters
MPEG-1 Audio 1992	Three layers of perceptual audio coding (I, II and III).	Layer III became MP3; mono and stereo audio.
MPEG-2 Audio 1994	Extends MPEG-1: lower sampling rates and backward-compatible multichannel audio.	Early surround sound and digital broadcasting.
MPEG-2 AAC 1997	A new, more efficient perceptual coder, designed for multichannel audio.	AAC; not backward-compatible with MPEG-1 decoders.

KEY POINT MPEG-1 and MPEG-2 established the core foundations of modern perceptual audio coding.

MPEG standards: audio, metadata and frameworks

These standards address different parts of the multimedia ecosystem.

Standard	Main purpose	Role
MPEG-4 Audio **1998 onward**	A toolbox for audio: AAC, speech, scalable, parametric and synthetic coding.	Codes and represents audio.
MPEG-7 **2001**	Standardizes metadata describing multimedia content.	Search and indexing; not an audio codec.
MPEG-21 **2002 onward**	A framework for identifying, managing and exchanging multimedia resources.	Interoperability and rights; not an audio codec.

KEY POINT MPEG-4 includes audio codecs; MPEG-7 and MPEG-21 do not compress audio.

MPEG-1 Audio

Lossy compression for mono and stereo signals.

- Sampling frequencies: 32 kHz, 44.1 kHz and 48 kHz.
- Precision: 16 bits.
- Three compression layers:
  - Layer 1: 32–448 kb/s, target 192 kb/s
  - Layer 2: 32–384 kb/s, target 128 kb/s
  - Layer 3: 32–320 kb/s, target 64 kb/s
- Layer 3 is the MP3 format, widely used for Internet applications.

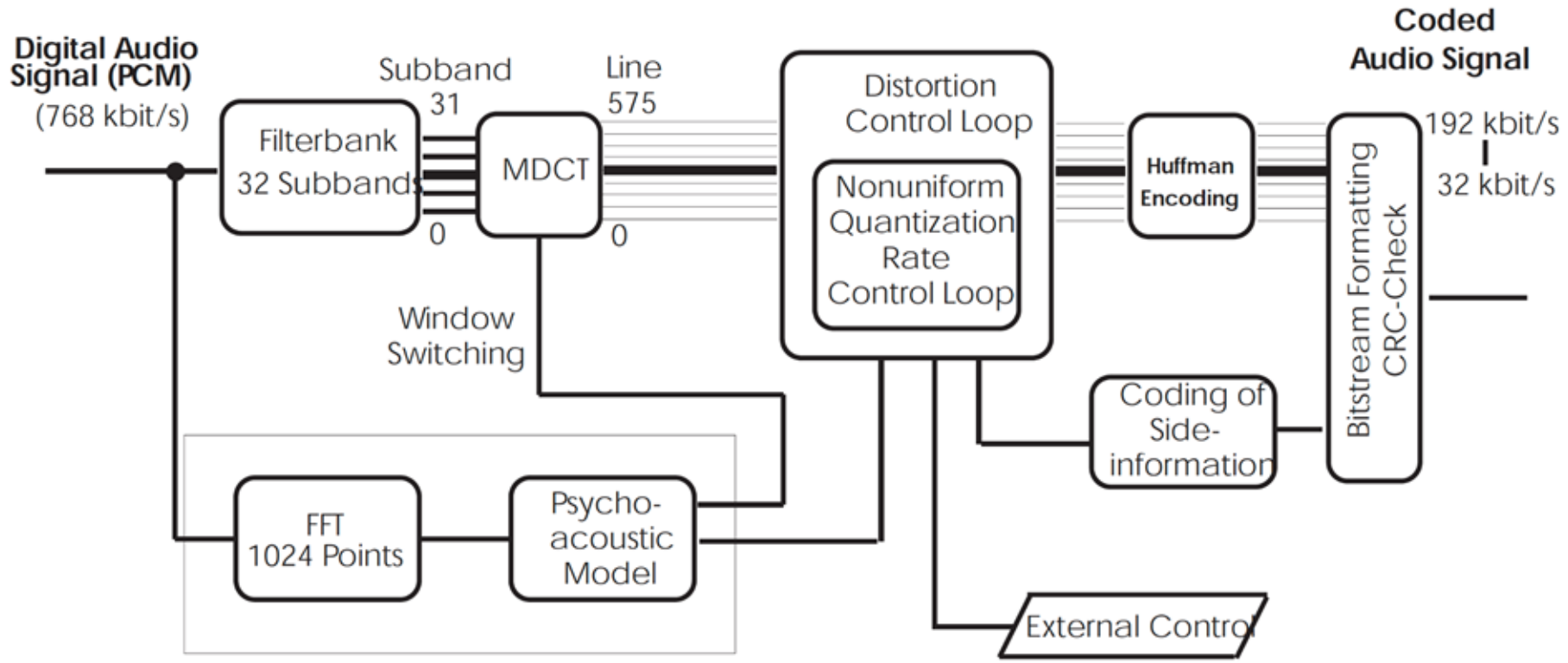
Layer 3 = MP3

Same family, increasing compression complexity from Layer 1 to Layer 3.

MP3 IN PRACTICE Bandcamp MP3 downloads (V0 / 320) · MP3 podcast episodes

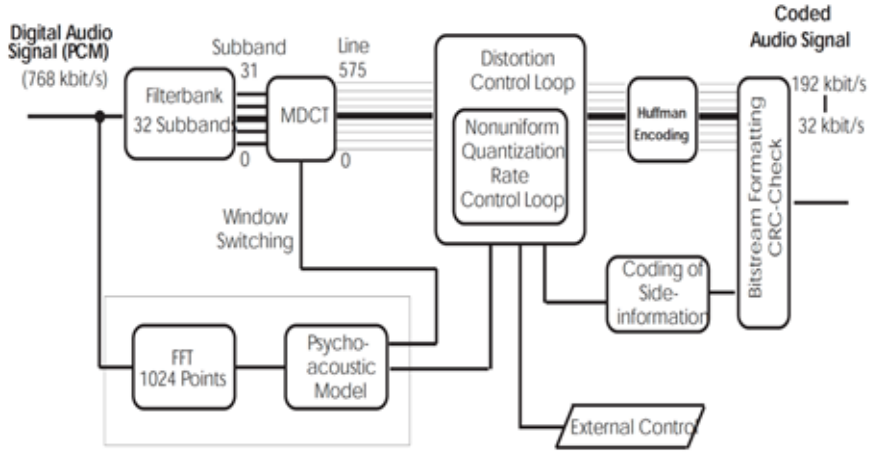
MP3 encoder

Hybrid filter bank, psychoacoustic model, nested rate/distortion loops and Huffman coding.



MP3 encoder: main components

- Filter bank: hybrid filter bank — polyphase filter bank followed by MDCT.
- Perceptual model: estimates masking thresholds using FFT and critical bands.
- Quantization and coding: two nested iteration loops and Huffman coding.



The standard fixes the decoder/bitstream; encoder quality depends on model and rate-control decisions.

MP3 audio format and bitstream fields

An MP3 file is a sequence of frames, often with metadata and per-frame headers.

An MP3 File



ternal Structure of an MP3 File

MP3 Header
MP3 Data
MP3 Header
MP3 Data
+++ Repeated +++
MP3 Header
MP3 Data
MP3 Header
MP3 Data

Note that the MP3 file structure may be 'encapsulated' within an ID3 tag.

ID3v2x Metadata
MP3 Header
MP3 Data
MP3 Header
MP3 Data
+++ Repeated +++
MP3 Header
MP3 Data
MP3 Header
MP3 Data

An MP3 Frame

MP3 Header
MP3 Data

Example MP3 Header

FFFBA040

Colour-coding shows binary bit mapping to hex values below

Detail of an MP3 Header

Bits	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
Binary	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0
Hex	F			F			F						B				A				0				4				0			
Meaning	MP3 Sync Word												Version	Layer		Error Protection	Bit Rate				Frequency		Pad. Bit	Priv. Bit	Mode		Mode Extension (Used With Joint Stereo)		Copy	Original	Emphasis	
Value	Sync Word												1 = MPEG	01 = Layer 3		1 = No	1010 = 160				00 = 44100 Hz		0 = Frame is not padded	Unknown	01 = Joint Stereo		0 = Intensity Stereo Off	0 = MS Stereo Off	0 = Not Copy-righted	0 = Copy Of Original Media	00 = None	

MPEG-2 Advanced Audio Coding (AAC)

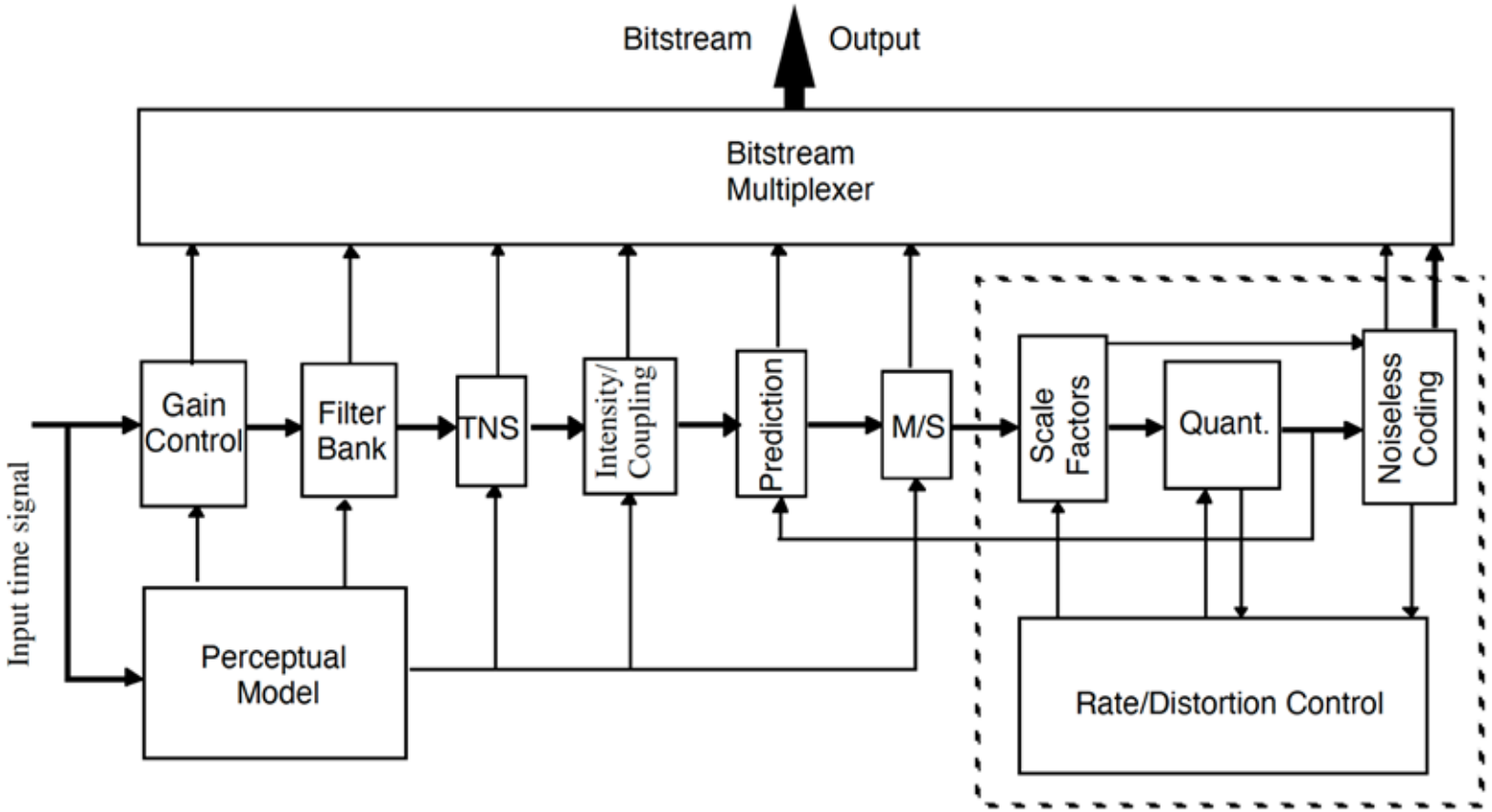
AAC follows the Layer-3 paradigm but adds tools for improved low-bitrate quality.

- Supports mono/stereo and multichannel audio (up to 5.1 in MPEG-2 AAC).
- Transform-based perceptual coding: MDCT, non-uniform quantization, Huffman coding and rate–distortion control.
- Compared with MP3: higher frequency resolution (1024 lines), improved joint stereo and block switching.
- Additional tools include temporal noise shaping, prediction and channel coupling.

AAC IN PRACTICE Spotify Web Player (AAC 128 / 256 kb/s) · Bandcamp AAC downloads

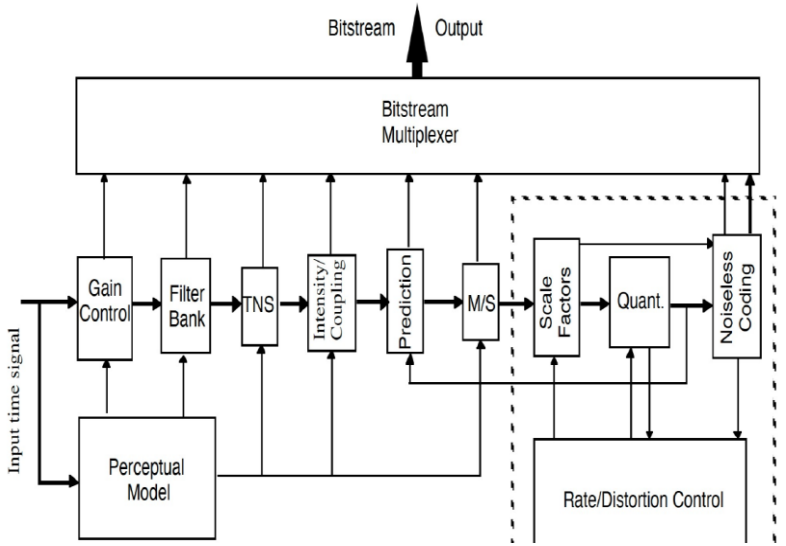
AAC encoding

AAC extends the perceptual transform-coding architecture with additional coding tools.



AAC coding tools

- Temporal Noise Shaping (TNS): controls the temporal shape of quantization noise.
- Intensity coding and coupling: reduce perceptually irrelevant high-frequency channel information.
- Prediction: removes redundancies between adjacent frames.
- M/S coding: removes stereo redundancy by coding sum and difference rather than left and right.



AAC improves MP3 mainly by adding better tools around the same core transform-coding idea.

MPEG-4 Audio: a toolbox, not a single codec

Part of MPEG-4 (ISO/IEC 14496-3); different coding tools address different signals and needs.

General Audio

AAC and related transform-domain perceptual coding tools for music and mixed audio.

Speech coding

Specialized predictive approaches (e.g. CELP) for speech at low bitrates.

Other MPEG-4 tools

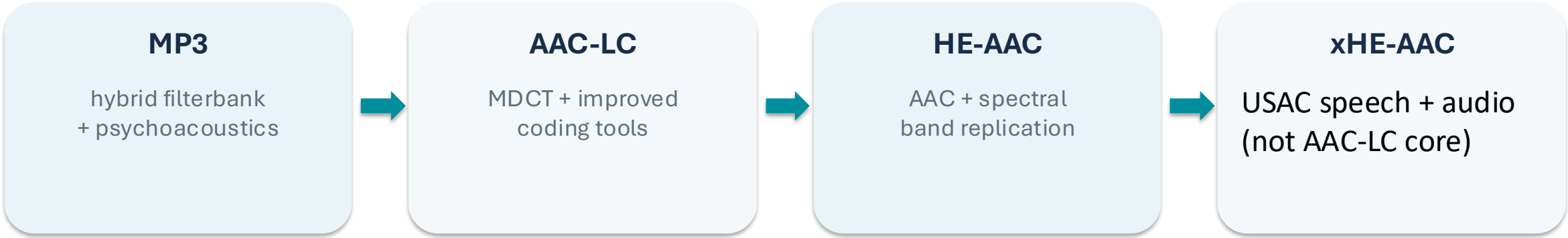
Scalable coding, parametric/synthetic audio and older experimental profiles.

Historical context TwinVQ and Structured Audio are useful examples, but are not central to today's deployed codecs.

MPEG-4 IN PRACTICE AAC audio in .m4a music files and .mp4 video

AAC evolution—and an independent alternative

AAC-LC → HE-AAC → xHE-AAC: successive solutions for more demanding low-bitrate use cases.



Parallel path Opus (SILK + CELT) separately unifies speech and music, prioritizing very low latency and packet-loss resilience.

HE-AAC: make high frequencies cheaper

HE-AAC v1 = AAC-LC + SBR | HE-AAC v2 = AAC-LC + SBR + Parametric Stereo

AAC-LC core

Transmits the lower-frequency audio using conventional perceptual transform coding.

SBR extension

Sends compact parameters to reconstruct high-frequency content at the decoder.

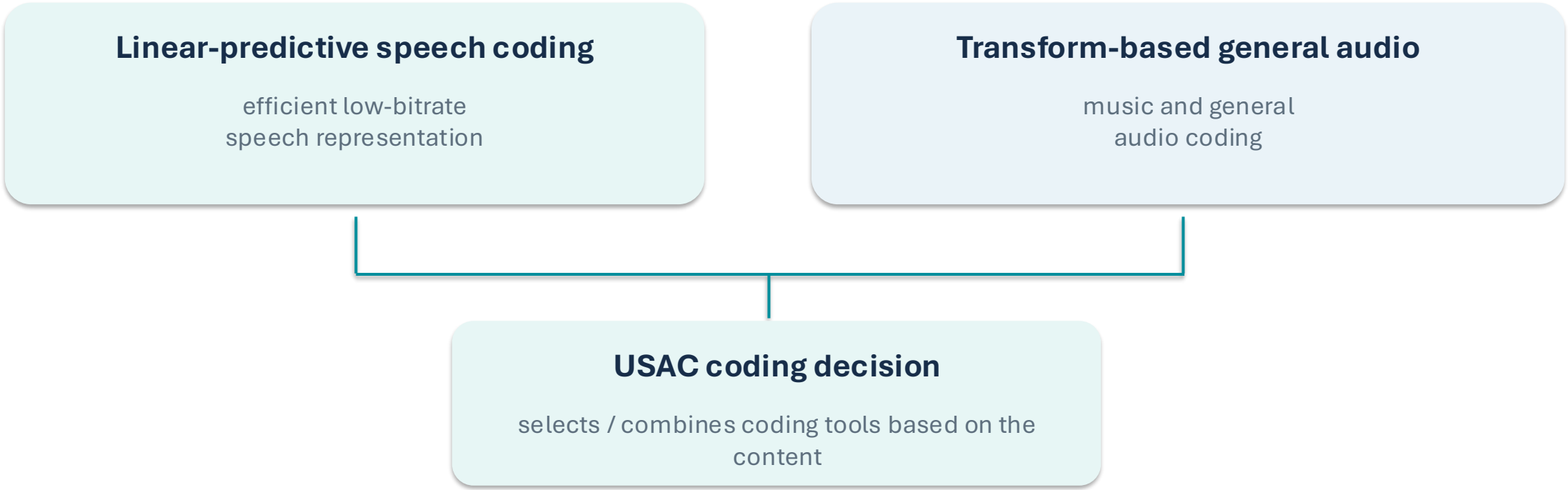
HE-AAC v2 adds Parametric Stereo: compact spatial parameters instead of fully coding two channels.

Best suited to bandwidth-constrained delivery; these tools trade some reconstruction fidelity for lower bitrate.

IN PRACTICE **DAB+ digital radio · low-bitrate Internet radio**

xHE-AAC: unified speech and audio coding

MPEG-D USAC: a new coding architecture, not AAC-LC plus another extension.



Wide bitrate range · speech/music switching · MPEG-D DRC and loudness metadata

xHE-AAC IN PRACTICE Netflix audio on supported Android devices · Android xHE-AAC decoding

Opus: one codec for speech and music

IETF RFC 6716 · especially relevant to interactive communication and low latency.

SILK

Predictive / SILK

Linear-prediction-based coding, optimized for speech.

CELT

Transform / CELT

MDCT-based coding, suited to music and low delay.



Adaptive operating modes

- SILK-only
- CELT-only
- Hybrid SILK + CELT

Practical traits: variable bitrate · adaptable frame duration · packet-loss concealment · WebRTC

OPUS IN PRACTICE Discord voice chat · WebRTC calls in Chrome / Firefox / Safari

Audio codecs: different optimization goals

Compare coding principle, low-bitrate efficiency, latency and deployment—not one universal winner.

Codec	Core strategy	Strength	Example uses
MP3	hybrid transform	legacy compatibility	downloads / archives
AAC-LC	MDCT perceptual	general-purpose audio	video / streaming
Opus	SILK / CELT / hybrid	low delay + versatility	calls / WebRTC
xHE-AAC	USAC: LP + transform	very low to high rates	adaptive delivery
HE-AAC v1/v2	AAC-LC + SBR / PS	efficient low-rate stereo	digital radio / streaming

Dolby AC-3 (Dolby Digital)

A de facto standard for movie theaters and home theater systems.

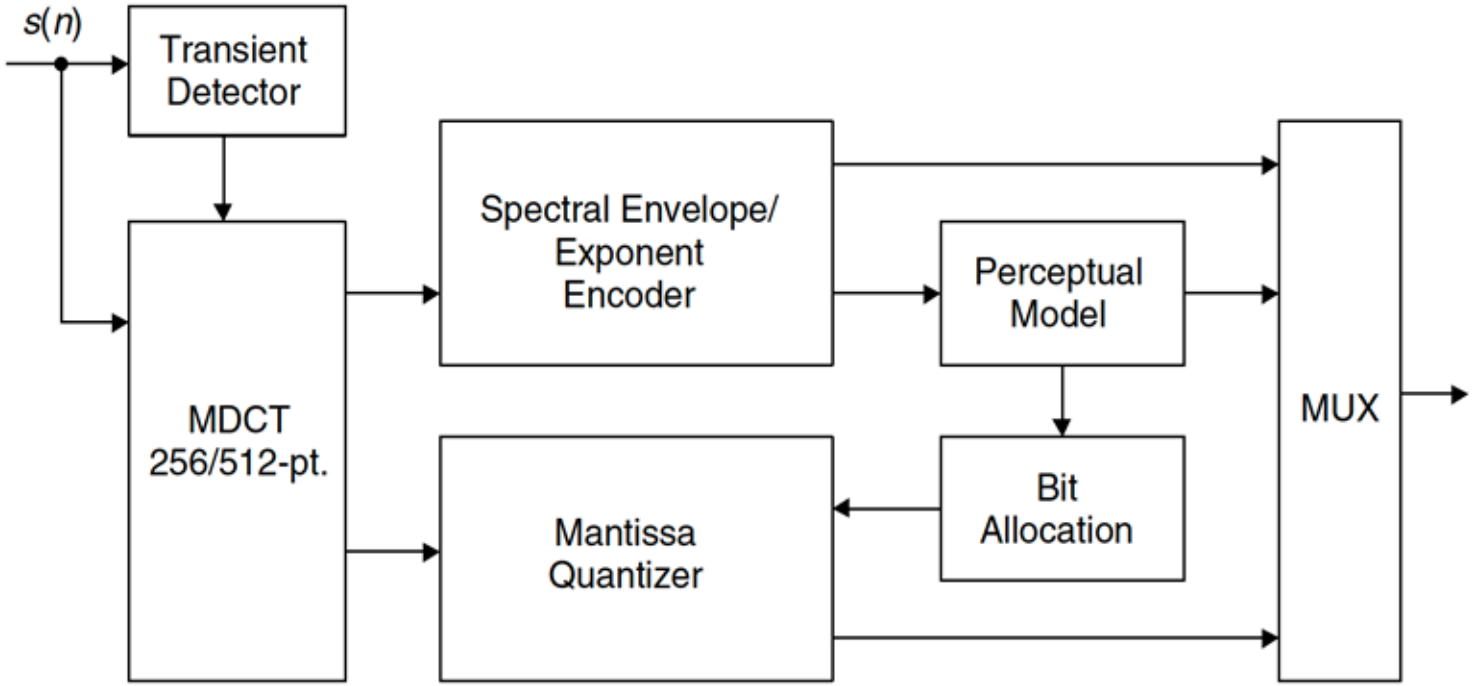
- Sample rates: 32, 44.1, and 48 kHz
- Bit rates: 32–640 kb/s, variable
- High-quality output at 64 kb/s per channel
- Delay roughly 100 ms
- MDCT filter bank
- Multiple channels processed as an ensemble
- Robust decoder downmix functionality from 5.1 to fewer channels

AC-3 was designed around multichannel playback and robust consumer decoding.

AC-3 IN PRACTICE DVD-Video surround tracks · digital TV and home-theater receivers

Dolby Digital encoder

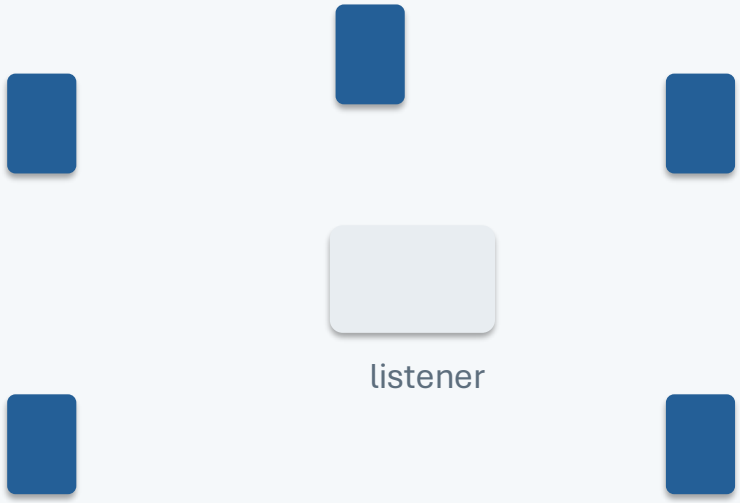
MDCT, transient detection, perceptual model, bit allocation and mantissa quantization are combined before multiplexing.



Channel-based versus object-based audio

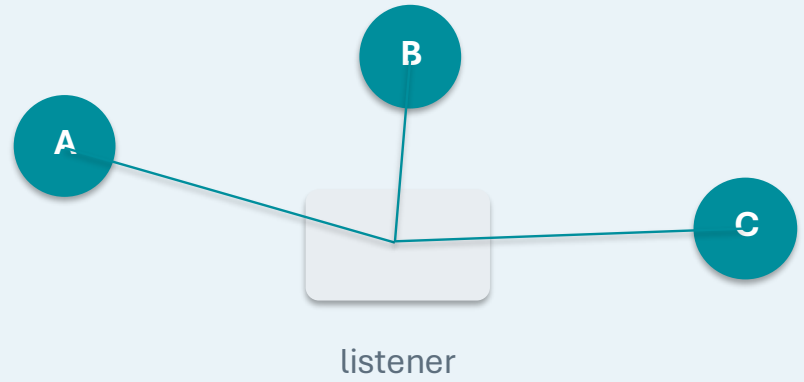
A new representation of the sound scene—not simply another compression algorithm.

Channel-based: 5.1 / 7.1



Signals assigned to fixed speaker channels

Object-based: Atmos

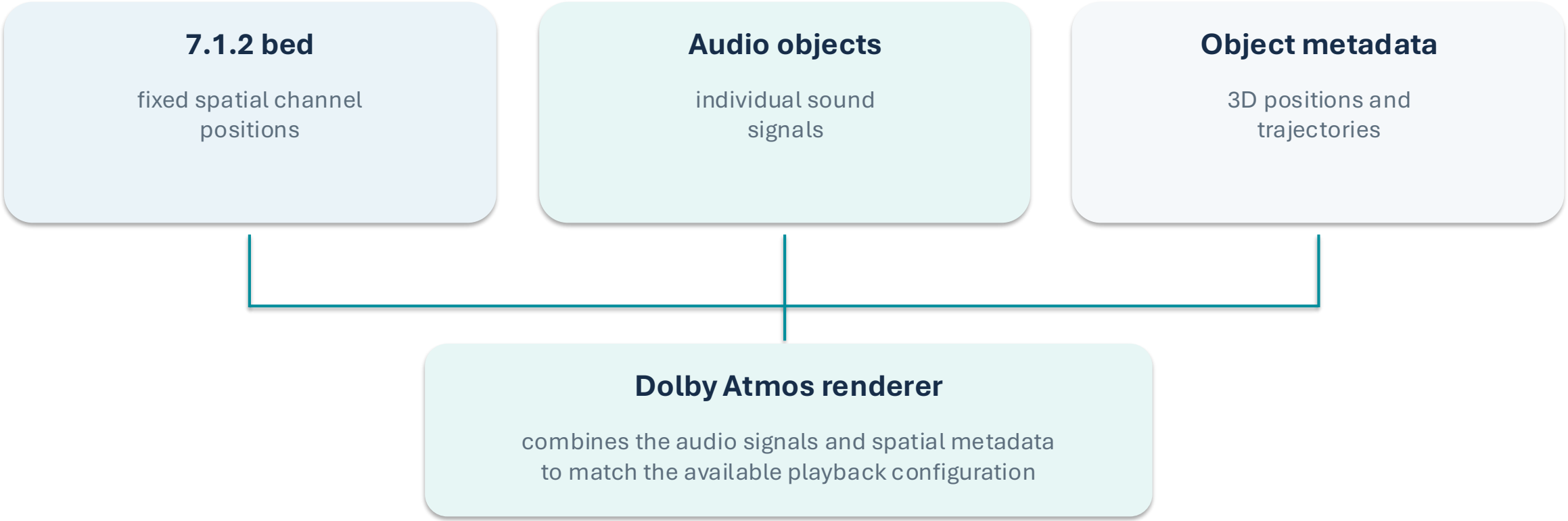


Audio signal + time-varying (x, y, z) metadata

ATMOS IN PRACTICE Apple Music Spatial Audio · Netflix Dolby Atmos titles · compatible soundbars

Dolby Atmos: beds, objects and metadata

In production, a fixed-channel bed coexists with spatially placed audio objects.



Up to 128 audio inputs in production (bed channels + objects); delivery formats may use fewer decoded elements.

Atmos: from production to playback

Rendering happens according to the available speakers or headphone configuration.

Mix / master

beds + objects
+ metadata

Distribution examples

- DD+ JOC: streaming (lossy)
- TrueHD: Blu-ray (lossless)
- AC-4: selected delivery systems

Decoder + renderer

spatial mapping depends on the device

Atmos describes the scene; transport codecs determine how efficiently it is delivered.

Cinema large speaker arrays

Home soundbar / AVR

Headphones binaural rendering

Dolby Digital Plus JOC: efficient Atmos delivery

Joint Object Coding provides backward-compatible object-based playback.

Atmos production mix

beds, objects,
spatial metadata

DD+ JOC bitstream

multichannel render
+ JOC side information

Regular DD+ decoder

outputs channel mix

Atmos-aware decoder

JOC reconstruction +
metadata for renderer

Important distinction

JOC delivers up to 15 full-range channels/objects plus LFE, rather than separately coding every production object.

MPEG-H 3D Audio: immersive and interactive

ISO/IEC 23008-3 · standardized audio coding, scene description and reproduction.

Channels

conventional beds

Objects

signals + metadata

HOA

sound-field representation

Personalization under producer control

Choose a language or commentator · boost dialogue · render for speakers, soundbars or headphones

MPEG-H IN PRACTICE South Korean ATSC 3.0 television · Sony 360 Reality Audio format

Immersive audio: what to compare

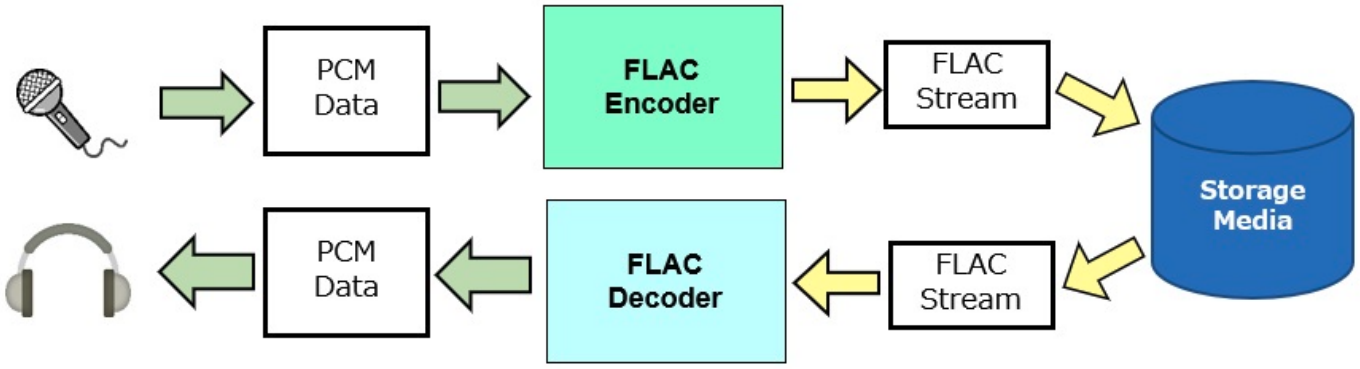
Compare representation, coding, rendering and deployment separately.

	Traditional 5.1	Dolby Atmos	MPEG-H 3D Audio
Representation	fixed channels	beds + objects	channels / objects / HOA
Spatial playback	fixed speaker mapping	adaptive rendering	adaptive rendering
User interaction	limited	format/application-specific	personalization supported
Coding	AC-3 / AAC etc.	DD+ JOC / TrueHD etc.	MPEG-H codec

Lossless audio coding: FLAC

FLAC preserves the original PCM audio exactly while reducing file size.

- Lossless compression: decoded PCM is bit-for-bit identical to the original.
- Typically compresses music substantially, depending on the signal.
- Open, non-proprietary format with fast decoding and rich metadata.
- Real products: TIDAL High/Max FLAC streaming and Bandcamp FLAC downloads.



FLAC properties

- **Lossless:** The encoding of audio (PCM) data incurs no loss of information, and the decoded audio is bit-for-bit identical to what went into the encoder.
- **Fast:** FLAC is asymmetric in favor of decode speed. Decoding requires only integer arithmetic, and is much less compute-intensive than for most perceptual codecs.
- **Hardware support:** FLAC is supported by dozens of consumer electronic devices, from portable players, to home stereo equipment, to car stereo.
- **Flexible metadata:** FLAC's metadata system supports tags, cover art, seek tables, and cue sheets. Applications can write their own APPLICATION metadata once they register an ID.
- **Seekable:** FLAC supports fast sample-accurate seeking. Not only is this useful for playback, it makes FLAC files suitable for use in editing applications.
- **Streamable:** Each FLAC frame contains enough data to decode that frame.

FFmpeg: experiment with codec choices

An open-source toolkit for transcoding, probing and evaluating encoded audio.

Opus / interactive audio

```
ffmpeg -i input.wav -c:a libopus -b:a 96k output.opus
```

AAC / general-purpose audio

```
ffmpeg -i input.wav -c:a aac -b:a 192k output.m4a
```

FLAC / lossless

```
ffmpeg -i input.wav -c:a flac output.flac
```

What to remember

Audio coding now optimizes more than file size.

01 · Signal coding

MP3 and AAC compress PCM using perceptual models, transforms, quantization and entropy coding.

02 · Application needs

Opus targets low delay and robustness; xHE-AAC spans speech and music at diverse bitrates.

03 · Sound scenes

Atmos and MPEG-H describe spatial information; renderers adapt playback to the listener.

Main distinction: codec • spatial representation • rendering • delivery format

References and further reading

Original textbook + standards and technical documentation for the new material.

- Spanias, Painter & Atti (2007), Audio Signal Processing and Coding
- Dolby — Beds and Objects in Dolby Atmos
- IETF RFC 6716 — Opus Audio Codec
- Dolby — Dolby Digital Plus JOC
- MPEG-D USAC (xHE-AAC) — ISO/IEC 23003-3
- Fraunhofer IIS — MPEG-H Audio
- FFmpeg documentation
- Free Lossless Audio Codec (FLAC)